

Searching for Submersibles

Summary

Greek company MCMS plans to explore Ionian Sea shipwrecks with its submersibles, pending safety approvals. They require a model to track the submersible's position, accounting for underwater variables. Our goal is to optimize the submersible's deployment and search patterns, and to develop a model minimizing the time to lose track of its location.

We established four models: location prediction model, equipment selection model, search path model and generalized model.

In the **location prediction model**, we synthesized dynamics, environmental factors, and a disturbance-free framework to create a **realistic motion model** for the motion of the submersible. The model focuses on key factors that affect the position of the submersible, such as the bottom topography, fluctuating seawater density, and ocean currents. We and tested the model for sensitivity with an **error model** in **section 5.6**.

In the **equipment selection model**, we used the **Analytic Hierarchy Process (AHP)** and **cost-benefit analysis** to optimize the search **efficiency** for the submerged wreck while taking cost effectiveness into account. We evaluated our **five chosen pieces of equipment** using this method: diving hooks, lifeboats, multibeam sonar, CTD sensors, and Doppler velocimeters. **Six major factors** guided our analysis: Availability, Maintenance Requirements, Readiness, Cost, Durability, and Controllability. This thorough analysis ensured a balanced evaluation, integrating both performance and economic considerations for optimal equipment selection.

The **search path model** focused on developing an efficient submersible **search procedure that minimizes ground coverage and maximizes resource use**. This was achieved by organizing the search to minimize redundant pathways and optimize time and resource consumption. This strategy, utilizing data from **initial search sites** and **expanded matrix search**, ensures an efficient and productive search process.

The **generalized model** adapts to **localized ocean conditions**, including temperature, density, currents, and topography, to ensure a safe and enjoyable travel experience for tourists using submersibles. This customized approach provides specialized safety and navigation strategies for each destination, ensuring a safe and enjoyable experience.

Finally, we performed comprehensive **sensitivity and error analyses**, and our model showed a high degree of accuracy. Through these analyses, we verified the robustness of the model under a variety of conditions and assessed the impact of potential errors on the model outputs to ensure its applicability in real-world situations.

Keywords: Realistic Motion Model; AHP; Sensitivity Analysis; Expanded Matrix

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1 Introduction

1.1 Problem Background

The Greek company Maritime Cruises Mini-Submarines (MCMS) specializes in the production of submersibles capable of carrying people deep into the sea. They plan to carry out tourist expeditions in the Ionian Sea, using submersibles to explore the wrecks of sunken ships. But before they could do so, MCMS needed to develop safety procedures to deal with potential communication disruptions or mechanical failures, including propulsion system failures. Regulators require them to develop a model to predict the position of the submersible over time, taking into account the challenges of the underwater environment, such as the varying densities of seawater, currents, and seafloor topography. This model is critical to enhancing safety measures for these underwater tourism activities.

The Ionian Sea is a region of converging and interacting water masses. Through the Strait of Sicily in the west, the Levantine Sea in the southeast, the Aegean Sea in the east, and the Strait of Otranto in the north, the Adriatic Sea is connected to the western Mediterranean. The deep and bottom waters of the Ionian and Levantine basins, from which the Adriatic Deep Water (ADW) flows via the Strait of Otranto, are primarily and historically contributed by the southern Adriatic Sea, which is the traditional "engine" of the internal conveyor belt of the eastern Mediterranean.



Figure 1: Deep Ocean [7]

1.2 Restatement of the Problem

Over time, the location of a submersible may be affected by ocean currents, differences in ocean density and/or seafloor geography. The location of the submersible needs to be known by predicting changes in the ocean environment. In view of the background information, constraints outlined in the problem statement and additional guidance, the following issues need to be addressed:

- Taking many factors into consideration, such as seafloor topography, currents, ocean density, to the location of the submersible.
- Considering the availability, maintenance, readiness, and cost associated with the use of equipment, what additional equipment should be equipped on the

host ship and rescue vessels?

- Predict the optimal method of finding the submersible to reduce the time to lose the submersible.
- Predict the probability of finding a submersible.
- Changes to expand the model to other tourist destinations.

1.3 Our Work

This problem requires us to build a submersible position prediction model and reduce the time to locate a lost submersible. Our work mainly includes:

1. Establish a motion model for the submersible based on the dynamic basis, natural factors and the influence of advanced equipment.
2. Search efficiency is optimized by Analytic Hierarchy Process (AHP) and cost-benefit analysis.
3. Based on the expanded matrix model, the effectiveness and applicability of the algorithm are effectively demonstrated.

In order to avoid complicated description, intuitively reflect our work process, the flow chart is shown in Figure 2:

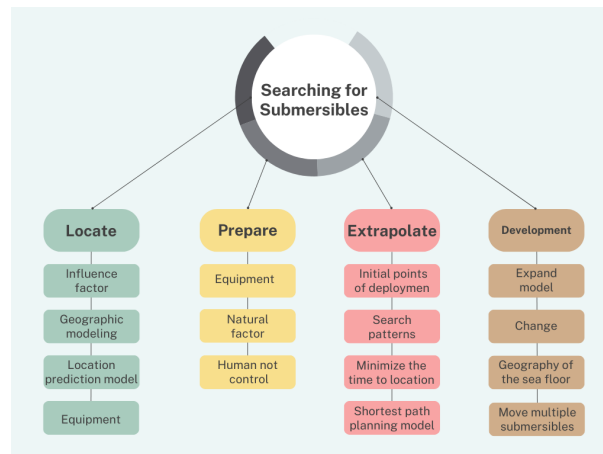


Figure 2: Our work

2 Assumptions and Explanations

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

Assumption 1: The submersible studied in this paper is a rigid body with left-right symmetric mass uniformly distributed.

Explanation: The assumption of a rigid body with symmetrical mass

distribution between the left and right sides of the submersible is motivated by the need for stability, predictable dynamics, efficient control and navigation, safety, and accurate data collection.

Assumption 2: The effect of wind is not considered

Explanation: Wind mainly affects the surface of the water column and has very little effect on deep diving submersibles.

Assumption 3: The effect of temperature is not considered

Explanation: In deeper water, water temperatures are relatively stable and do not vary much.

Assumption 4: Vibration and noise from the submersible itself are not considered.

Explanation: The submersible operates steadily and internal vibration or noise cannot affect the trajectory of the submersible.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
u	Velocity component
t	Time
v	Velocity
ρ	Density of sea water
V	Volume of submersible
F and f	Force
W and w	Weight
θ, ϕ, α	Angle
p	pressure
g	gravity
C	Total cost of purchasing equipment and maintaining it.
E	Total efficient of equipment.
B_k	Criteria for equipment
C_i	Selection for equipment

*There are some variables that are not listed here and will be discussed in detail in each section.

4 Model Preparation

4.1 Data Overview

The problem does not provide us with data directly, so we need to consider what data to collect when constructing the model. By analyzing the problem, we need to

collect information about the factors that affect the submersible and other factors, such as **the buoyancy of seawater, the density of seawater, ocean currents, and seafloor topography** and so on. By analyzing the problem, we need to collect information about the factors affecting the submersible as a percentage of each factor. We build an actual motion model of the submersible that integrates the motion model, environment model, and free perturbation model by collecting topographic maps of the relevant sea area, and combining the effects of seafloor topography on ocean currents, and the effects of ocean currents and seawater density on the submersible's motion. Due to the large amount of data, it is inconvenient to list them one by one, so creating a pairwise comparison matrix for display is a good method.

4.1.1 Data Collection

The official website of Gebco and ResearchGate was queried and lots of data about the Ionian Sea were obtained. And data sources are shown in Table 2.

Database Names	Database Websites
Gebco	https://www.gebco.net/
ResearchGate	https://www.researchgate.net/
Google Scholar	https://scholar.google.com/
Maps	© 2021 Mapbox © OpenStreetMap

Table 2: Data and Database Websites

4.1.2 Data Screening

The first image provides a bathymetric map of the Ionian Sea (part of the Mediterranean Sea), which includes various geological features and sampling points that can be used for scientific research, for example, to study underwater topography.

The second image, in turn, reveals the major current systems in the Ionian Sea, showing how bodies of water move within this sea area. The patterns of ocean currents have a profound effect on the marine environment, from regulating ocean temperature and salinity.

With the picture above, we can choose the area rationally when constructing the location model, because the main problem to be solved is the complex uncertainty, such as ocean currents, seafloor topography, seawater density and so on.

5 Location Prediction Model

In order to develop a more accurate model for predicting the position of a submersible. Our model aims to dynamically capture and predict the position of the submersible in these changing environments to optimize the efficiency of search and rescue operations. First of all, we assumed that the hydrodynamic coefficients and related parameters of the submersible are constant, and externally environmental changes will not cause them to change. Then, we focused on several **uncertainties** that affect the position of the submersible: **mechanical defects, ocean currents, regions of varying sea-**

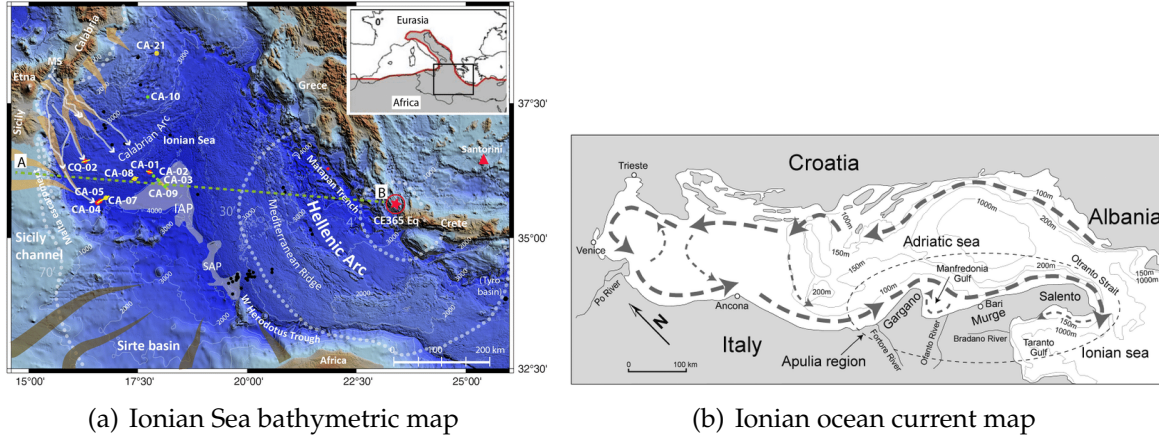


Figure 3: Data Screening

water density, and geographic features of the seafloor. Our model integrates and analyzes the complex relationship between the position of the submersible and the ocean currents, seawater temperature, and density. The submersible needs to be equipped with some advanced devices, such as gyroscopes, which provide **precise geographic information**, and we can predict the position of the submersible from the data provided by these devices. In addition, we have explored the effects of **natural factors** on submersibles, and we have focused on the interactions of submersibles with the seafloor topography, especially in complex and irregular terrain, which may significantly affect the navigational path and stability of submersibles. We then discuss the motion patterns of the submersible when **mechanical defects** are encountered. Finally, we perform an **error analysis**.

5.1 Predicting Submersible Positions Using Kinematics

The forces on the submersible in the water include hydrodynamic forces, hydrodynamic moments, and its own forces and moments [10], and we can get the following dynamic equations of the submersible:

$$M_{RB}\dot{V} + M_A\dot{V} + C_A(V)V + C_{RB}(V)V + D(V) + g(\eta) = \tau + \omega \quad (1)$$

In the above equation, M_{RB} represents the mass matrix of the submersible, M_A represents the added mass matrix of the submersible, V represents the velocity and angular velocity vector of the submersible in the body-fixed coordinate system, $C_{RB}(V)$ represents the submersible's Coriolis matrix, $C_A(V)$ represents the hydrodynamic Coriolis matrix, $D(V)$ represents the nonlinear fluid dynamic damping matrix, τ is the submersible's input control matrix, and ω represents environmental disturbances.

The damping matrix $D(V)$ is decomposed into the following components:

$$D(V)V = D_l V + u D_u V + D_{nl}(V)V \quad (2)$$

5.2 Uncertainties

The location of a submersible may be affected by environmental factors such as: ocean currents, varying densities of the ocean and the geography of the seafloor. De-

fective submersibles (including submersibles that have lost propulsion) are likely to be placed on the seafloor or at some neutral buoyancy point underwater.

5.3 Equipment and Effect

We consider that submersibles need to be equipped with advanced equipment such as sonar, inertial navigation systems, CTD sensors, gyroscopes, Doppler tachometers, multibeam sonar, etc. We use these equipment to obtain more accurate data: such as the position of the submersible, seawater density, inclination of the submersible, speed of the submersible, seafloor topography, etc., to understand and predict the dynamics of the submersible, and to send information to the host ship on a regular basis to reduce uncertainty [5]. Submersibles equipped with advanced equipment can provide critical information to the host ship: precise position data, seawater density, seawater velocity, orientation and attitude information, and seafloor topography information [6] [8]. We believe that it is important to understand the dynamics of the surrounding environment and predict the position of the submersible by installing advanced technology on board the submersible, performing precise geolocation and measuring the flow rate and density of seawater. This information is critical to monitoring the status of the submersible, improving safety and optimizing navigational efficiency, which together ensure that the submersible operates safely and efficiently in an intricate underwater environment.

- 1) **Sonar systems** detect underwater objects by emitting and listening for sound waves, where active sonar emits sound waves and receives echoes to find targets, passive sonar listens for potential noise, and side-scanning sonar generates high-resolution images of the seafloor for large-area searches [9].

D is the distance between the submersible and the object

T is the time interval between emission and reception of sound waves

V is the speed of propagation of the sound wave through the water

$$D = \frac{V}{T} \quad (3)$$

The inertial navigation system calculates the changes in position, velocity, and direction of the submersible by measuring its acceleration a and angular velocity.

$$\vec{p}(t) = \vec{p}(t_0) + \int_{t_0}^t \vec{v}(r) dr \quad (4)$$

$$\vec{v}(t) = \vec{v}(t_0) + \int_{t_0}^t \vec{a} dr \quad (5)$$

$p(t)$ is the position vector of t , v is the velocity vector, a is the acceleration vector, and T is any moment t_0 to t .

- 2) **CTD sensors** provide us with the physical properties of the ocean, such as density, by measuring the conductivity, temperature and depth of the water [3].

$$\rho(s, T, P) = \rho_0 + \delta\rho(s, T, P) \quad (6)$$

- 3) The **gyroscope** is able to measure the angular velocity of the submersible relative to its axis of rotation [2], allowing the inclination of the submersible to be calculated.

The rotation vector can be expressed as $\vec{\omega} = (\omega_x, \omega_y, \omega_z)$, and the differential of the rotation angle vector can be expressed as $\Delta\vec{\theta} = (\Delta\theta_x, \Delta\theta_y, \Delta\theta_z)$, then the angle of inclination of the submersible can be derived as:

$$\Delta\vec{\theta} = \vec{\omega}\Delta t \quad (7)$$

$$\vec{\theta}(t) = \vec{\theta}(t_0) + \int_{t_0}^t \vec{\omega}(r) dr \quad (8)$$

- 4) **Doppler velocimeters** are able to accurately measure velocity using the Doppler effect, and are particularly suited to detecting the rate of a moving object relative to a detector, so we use Doppler velocimeters to measure the velocity of submersibles.

$$f' = f \left(\frac{v + v_d}{v + v_s} \right) \quad (9)$$

f' is the reflected frequency, f is the emitted frequency, and v is the propagation speed of the acoustic wave in the medium. v_d is the speed of movement of the DVL relative to the medium, v_s the speed of movement of the acoustic wave source relative to the medium.

- 5) **Multibeam sonar** can avoid obstacles by scanning the terrain in the forward direction, with a large measuring range, fast speed and high accuracy, expanding the range of measurement from the point and line of single-beam sonar to the surface, and the multibeam sonar is able to obtain a large range of detailed undersea terrain information [4].

d denotes the distance from the submersible to the seafloor, c is the speed of the submersible in the water, and the submersible is located at the position (x_0, y_0, z_0) and transmit sound waves with angle θ .

$$t = \frac{2d}{c} \quad (10)$$

$$x = x_0 + d \sin(\theta) \cos(\phi) \quad (11)$$

$$y = y_0 + d \sin(\theta) \sin(\phi) \quad (12)$$

$$z = z_0 - d \cos(\theta) \quad (13)$$

5.4 The Influence of Natural Factors

The position of the submersible may be affected by ocean currents, the varying densities of the ocean and the geography of the seafloor [10].

1) The speed of **ocean currents** is affected by the depth of water [1].

$$v(h) = v_0 \left(1 + d \left(\frac{h_0(x, y)}{h_0} \right) \right) \quad (14)$$

h_0 is the reference water depth and $h(x, y)$ is the water depth at the given position, d is the adjustment factor.

$$\rho = P(T, P, s) = \frac{\rho(T, O, s)}{1 - K_T(T, s, P)} \quad (15)$$

$$x = V_i t \cos(\theta) - \frac{-C_d \cdot A_x \cdot \rho \cdot (V_i \cdot \cos(\theta))^2 + F_c}{2m} t^2 \quad (16)$$

$$y = V_i t \sin(\theta) + \frac{C_d \cdot A_y \cdot \rho \cdot (V_i \cdot \sin(\theta))^2}{2m} t^2 \quad (17)$$

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho} \nabla p + \nu \nabla^2 u + f \quad (18)$$

$$\nabla \cdot u = 0 \quad (19)$$

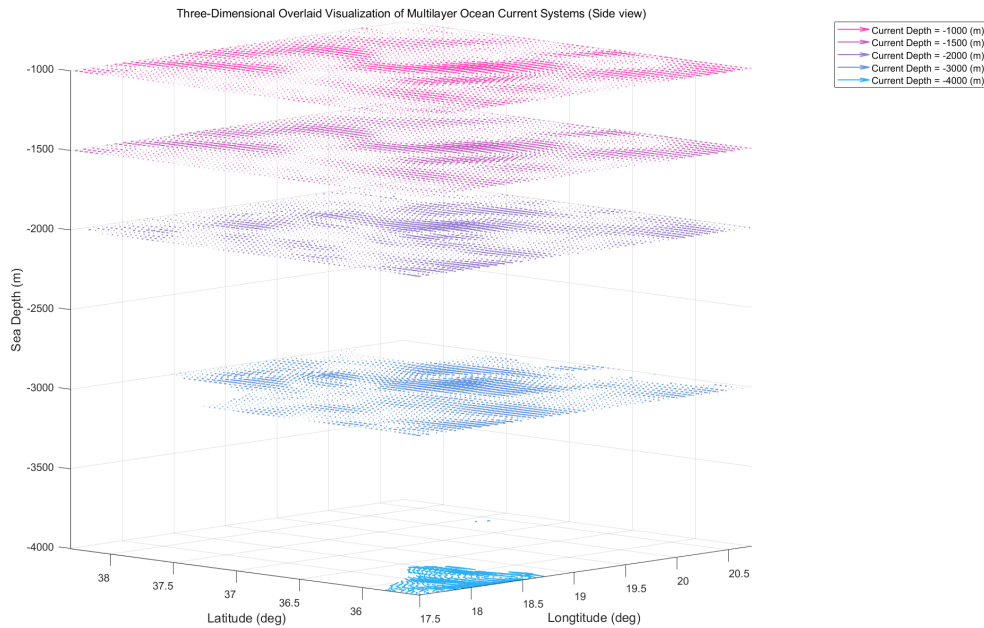


Figure 4: Three-Dimensional Overlaid Visualization of Multilayer Ocean Current

2) **Upslope seabed topography, downslope seabed topography, convex seabed topography, concave seabed topography** are some typical seabed topography, and they are also the common seabed topography encountered by submersibles when performing seabed target searching missions. These topographies are able to test the tracking capability of submersibles when encountering different topographies

on the seabed, and we have built corresponding mathematical expressions for different seabed topographies according to their characteristics.

Upslope seafloor terrain:

According to the terrain characteristics of the upslope seafloor terrain, we establish the following mathematical expression for it:

$$h = \begin{cases} h_0, & x \leq x_1 \\ h_0 + 0.5h_1 \left(1 + \tanh\left(3 \cdot 6 \frac{x-x_1-\frac{s_1}{2}}{s_1}\right) \right), & x_1 < x \leq x_1 + s_1 \\ h_0 + h_1, & x > x_1 + s_1 \end{cases} \quad (20)$$

In equation (20), h_0 represents the situation without control, h represents the situation after control relative to before control, x represents the aforementioned initial abscissa, s_1 represents the difference in abscissa at the end of the uphill section from the aforementioned initial abscissa, that is, the distance of the uphill journey.

Downslope seafloor terrain:

According to the characteristics of downslope seafloor terrain, we establish the following mathematical expression for it:

$$h = \begin{cases} h_2, & x \leq x_2 \\ h_2 - 0.5h_3 \left(1 + \tanh\left(3 \cdot 6 \frac{(x-x_2-\frac{s_2}{2})}{s_2}\right) \right), & x_2 < x \leq x_2 + s_2 \\ h_2, & x > x_2 + s_2 \end{cases} \quad (21)$$

In equation (21), h_2 represents the situation without control, h_3 represents the situation after control relative to before control, x_2 represents the starting abscissa of the descent, s_2 represents the difference in abscissa at the end of the downhill section from the starting abscissa, that is, the distance of the downhill journey.

Convex seafloor terrain:

Based on the characteristics of convex seafloor terrains, we have established mathematical expressions for them as follows:

$$h = \begin{cases} h_4, & x \leq x_3 \\ h_4 + 0.5h_5 \left(1 + \tanh\left(3 \cdot 6 \cdot \left(\frac{x-x_1-\frac{s_3}{2}}{s_3}\right)\right) \right), & x_3 < x \leq x_3 + s_3 \\ h_4 + 0.5h_5 \left(1 + \tanh\left(3 \cdot 6 \cdot \left(\frac{x_3+s_3-x_1-\frac{s_3}{2}}{s_3}\right)\right) \right), & x_3 + s_3 < x \leq x_3 + s_3 + l \\ h_4 + 0.5h_5 \left(1 - \tanh\left(3 \cdot 6 \cdot \left(\frac{x-l-x_1-\frac{s_3}{2}}{s_3}\right)\right) \right), & x_3 + s_3 + l < x \leq x_3 + 2s_3 + v \\ h_4, & x > x_3 + 3s_3 + l \end{cases} \quad (22)$$

In equation (22), h_4 represents the final height in the case of uncontrolled coasting, h_5 represents the final height of controlled coasting relative to uncontrolled coasting, x_3 represents the starting abscissa of the coasting, s_3 represents the difference in abscissa at the end of the coasting from the starting abscissa, that is, the distance of the coasting.

Concave seafloor terrain:

Based on the characteristics of concave seafloor terrains, the following mathematical expressions have been established for the concave seafloor terrains.

$$h = \begin{cases} h_6, & x \leq x_4 \\ h_6 + 0.5h_7 \left(1 - \tanh\left(3 \cdot 6 \cdot \frac{(x-x_1-\frac{s_4}{2})}{s_4}\right) \right), & x_4 < x \leq x_4 + s_4 \\ h_6 + 0.5h_7 \left(1 - \tanh\left(3 \cdot 6 \cdot \frac{(x_4+s_4-x_1-\frac{s_4}{2})}{s_4}\right) \right), & x_4 + s_4 < x \leq x_4 + s_4 + l \\ h_6 + 0.5h_7 \left(1 + \tanh\left(3 \cdot 6 \cdot \frac{(x-l-x_1-\frac{s_4}{2})}{s_4}\right) \right), & x_4 + s_4 + l < x \leq x_4 + 2s_4 + l \\ h_6, & x > x_4 + 2s_4 + l \end{cases} \quad (23)$$

In equation (23), h_6 represents the height in the case of uncontrolled coasting, h_7 represents the height of controlled coasting relative to uncontrolled coasting, x_4 represents the starting abscissa of the coasting, s_4 represents the difference in abscissa at the end of the coasting from the starting abscissa, that is, the distance of the coasting.

5.5 Defective Submersible

When the submersible has a mechanical failure, it loses propulsion, as a result, we consider the vertical terminal velocity and horizontal terminal velocity of the submersible as Figure 6.

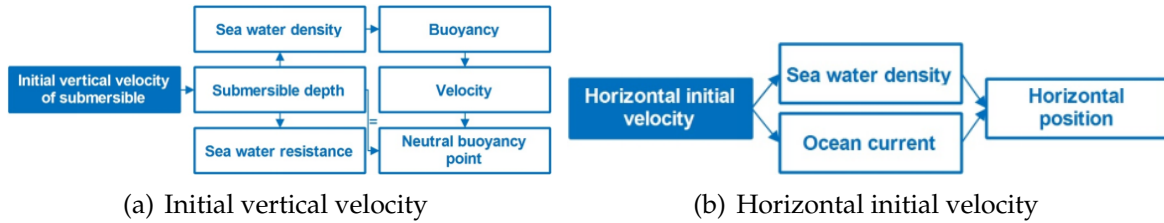


Figure 5: Velocity

- 1) This is the Navier-Stokes equation for **incompressible flow**.

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u + f \quad (24)$$

- 2) This is the equation for **incompressible flow**. Divergence of the velocity field equals zero for incompressible flow.

$$\nabla \cdot u = 0 \quad (25)$$

- 3) This is an equation for **velocity**. v_r is radial velocity, v_i is initial velocity.

$$v_r = v_i \cdot \sin(\alpha') \quad (26)$$

- 4) This is the **drag force** equation.

$$F_{\text{drag}}(u, v) = C_d \cdot A \cdot \rho \cdot V^2 \quad (27)$$

This equation is **weight** for buoyancy.

$$W = \rho \cdot V \cdot g \quad (28)$$

This equation is **typically** for buoyancy.

$$B = \rho \cdot V \cdot g \quad (29)$$

This states that **weight is equal to buoyancy**. B is Buoyancy force.

$$W = B \quad (30)$$

5) This is the **net force** equation

$$F_{\text{net}} = B - W - F_{\text{drag}} \quad (31)$$

$$a = \frac{F_{\text{net}}}{m} \quad (32)$$

$$\rho = \frac{s + 1305 - 0.3t + 5.25}{1.305} \quad (33)$$

6) This is the final horizontal velocity.

$$m \frac{dv}{dt} = -C_d \cdot A \cdot \rho \cdot V_i \cdot \cos^2(\alpha) + F \quad (34)$$

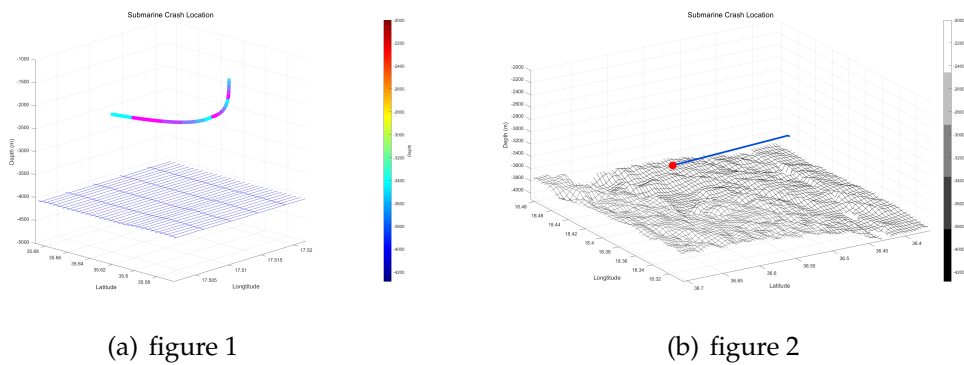


Figure 6: Submarine Crash Location

5.6 Error Model

In order to get the error between the actual motion trajectory and the expected motion trajectory of the submersible, we establish a path tracking error model.

Let u, v denote the surge and sway velocities of the submersible in the body frame, the motion equation in the horizontal plane can be expressed as follows: Let $v_t = \sqrt{u^2 + v^2}$; P is the position vector in the expected motion trajectory of the submersible; the angle β is the angle between the surge velocity u and the velocity v_t , which is given by: $\cos \beta = u/v_t$.

ψ is the bow angle, ψ_w is the angle of combined velocity v in inertial coordinate system ξ -axis in the positive direction, and ψ_e is angle of the tangent line of v_t through point P and ψ_F is the angle between the tangent line and the positive direction of the ξ -axis. The center point Q of submersible in the inertial coordinate system I $(x, y, 0)$, in the Serret-Frenet system F is $(x_e, y_e, 0)$.

The horizontal equation of motion of the submersible is:

$$\begin{cases} \dot{x} &= u \cos \psi - v \sin \psi \\ \dot{y} &= u \sin \psi + v \cos \psi \\ \dot{\psi} &= r \end{cases} \quad (35)$$

Let R denote the position vector in the Serret-Frenet system of the curve F , then R is:

$$R = \begin{bmatrix} \cos \psi_r & \sin \psi_r & 0 \\ -\sin \psi_r & \cos \psi_r & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (36)$$

$$(37)$$

The time derivative of yaw ψ is denoted by $\dot{\psi} = \psi_r$, where:

$$\begin{cases} \dot{\omega}_F = \dot{\psi}_F = c_c(s)\dot{s} \\ \dot{c}_c(s) = g_c(s)\dot{s} \end{cases}$$

Given that $c_c(s)$ and $g_c(s) = \frac{dc_c(s)}{ds}$, the derivative of the position vector P in the Serret-Frenet frame of the curve F can be expressed as:

$$\frac{dp}{dt} = \begin{bmatrix} \dot{s} \\ 0 \\ 0 \end{bmatrix} \quad (38)$$

According to equation (35), the speed in the inertial coordinate system $\{I\}$ of the submersible is :

$$\left(\frac{dq}{dt}\right)_I = \frac{dp}{dt}_F + R^{-1} \left(\frac{dd}{dt}\right)_F + R^{-1}(\omega_F \times d) \quad (39)$$

As stated in equation (39), the vector d describes the curve Q in the inertial coordinate system $\{I\}$, and the UUV follows the curve R in the Serret-Frenet system $\{F\}$:

$$R \left(\frac{dq}{dt}\right)_I = \frac{dp}{dt}_F + \frac{dd}{dt}_F + \omega_F \times d \quad (40)$$

$$\left(\frac{dq}{dt}\right)_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ 0 \end{bmatrix}, \quad \left(\frac{dd}{dt}\right)_F = \begin{bmatrix} \dot{x}_e \\ \dot{y}_e \\ 0 \end{bmatrix}$$

$$\omega_F \times d = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi}_F = -c_c(s)\dot{s} \end{bmatrix} \times \begin{bmatrix} x_e \\ y_e \\ 0 \end{bmatrix} = \begin{bmatrix} -c_c(s)\dot{s}x_e \\ c_c(s)\dot{s}y_e \\ 0 \end{bmatrix}$$

Following the equations, we can calculate the equation for path tracking error:

$$\begin{cases} \dot{x}_e = -s'(1 - c_c v_e) + v_t \cos \psi_e \\ \dot{y}_e = -c_s v_e + v_t \sin \psi_e \\ \dot{\psi}_e = r + \beta - c_c s' \end{cases} \quad (41)$$

6 Equipment Selection Model

6.1 Equipment Selection

We recommended that the company deploy CTD sensors, Doppler Velocimeters, multibeam sonar, lifeboats, and diving hooks on the host vessel, which play an important role in the field of ocean exploration and safety at sea. CTD sensors provide measurements of key oceanographic parameters such as water temperature, salinity, and depth, which are critical for oceanographic research and climate change monitoring. Doppler Velocimeters accurately measure the velocity and direction of flow in the water column and are useful for navigational navigation and ocean dynamics research. Multibeam sonar, on the other hand, plays a key role in mapping seafloor topography, providing accurate data for ocean mapping, resource exploration and environmental monitoring. Lifeboats, a key piece of maritime safety equipment, provide life protection in emergencies and can be rapidly deployed in harsh conditions. And diving hook systems provide divers with the ability to breathe underwater for longer periods of time, making shallow water diving safer and more comfortable. This equipment supports a wide range of marine activities, from scientific research to recreational diving, and is an integral part of marine exploration and activity.

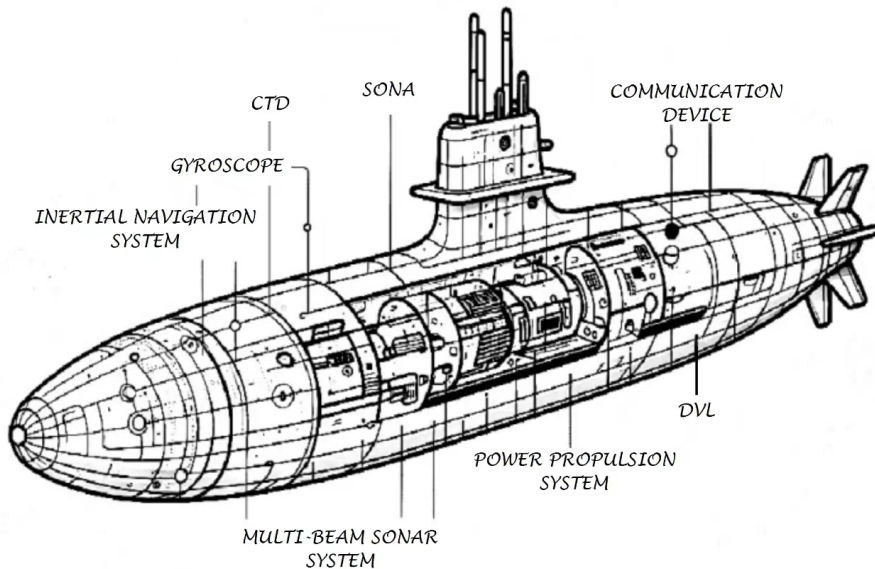


Figure 7: Schematic diagram of the submarine

6.2 Cost-benefit Analysis

We used a **cost-benefit analysis** to consider the costs associated with the availability, maintenance, readiness, and use of this equipment. Due to the complexity and

lack of quantification of the data, it consists of numerous interrelated and interlocking factors. The purpose of **CER (Cost Effectiveness Ratio)** is to determine the cost per unit of effect. E is the efficient value and C is the cost value. R denotes search and rescue range, A denotes accuracy, and D denotes reliability. w_1, w_2, w_3 are the weights of these factors.

$$CER = \frac{E}{C} \quad (42)$$

$$E = w_1 R + w_2 A + w_3 D \quad (43)$$

Then we use the **Analytic Hierarchy Process (AHP)** to apply to the decision-making problem by first structuring the problem and building a hierarchical model. Under this model, the complex problem of search equipment selection is decomposed into its constituent elements. These elements form several levels based on their attributes and relationships. The elements at the previous level provide guidance to the elements at the next level. These levels can be categorized into three types.

level 1: Top level

Explanation: There is only one element in this level, choice of search equipment is the intended goal or desired outcome of the analyzed problem.

level 2: Intermediate level

Explanation: This level contains the intermediate aspects involved in achieving the objectives, including the criteria to be considered: Availability, Maintenance Requirements, Readiness, Cost, Durability, Controllability.

level 3: Bottom level

Explanation: This level includes the various measures, decision-making options: CTD sensors, Doppler Velocimeters, multibeam sonar, lifeboats, and diving hooks.

Direct consideration of the extent to which each factor affects the factor often leads the decision maker to present data that is inconsistent with the actual level of significance that the decision maker believes to be due to incomplete consideration of the factors, so we have adopted a two-by-two comparison of the factors to create a pairwise comparison matrix. As to how to determine the value of a_{ij} , we refer to the numbers 1 to 9 and their reciprocals as a scale. Table 1 lists the meanings of the scales 1 to 9.

Table 3: Meaning of Scale

Scale	Meaning
1	Compared with two factors, it has the same importance.
3	Compared with two factors, the former is slightly more important than the latter
5	Compared with two factors, the former is significantly more important than the latter
7	Compared with two factors, the former is more important than the latter.
9	The former is extremely important than the latter.
2,4,6,8	Indicates the middle value of the above adjacent judgment.
reciprocal	The importance ratio of factor j to i, a_{ji} , is the reciprocal of factor i to j, $1/a_{ij}$.

we calculate a 6-order hierarchical single sorting vector from matrix A, which is generated as a consistency matrix. Since the eigenvalue is calculated by the inverse

method for a_{ij} , thus $\lambda_{\max}$ changes significantly from n , the degree of inconsistency of A becomes more severe, and the normalized eigenvector corresponding to $\lambda_{\max}$ cannot truly reflect the weight relationship $\mathbf{X} = \{x_1, \dots, x_6\}$. This affects the proportion of influence in the comparison matrix Z . Therefore, the consistency test of the judgment matrix provided by the comparison matrix is conducted to determine whether it can be accepted.

Table 4: Hierarchical single sorting matrix

A	B_1	B_2	B_3	B_4	B_5	B_6
B_1	1	7	5	1	3	1/3
B_2	1/7	1	1/6	1/7	1/5	1/7
B_3	1/5	6	1	1/5	1/4	1/5
B_4	1	7	5	1	3	1
B_5	1/3	5	4	1/3	1	1/5
B_6	3	7	5	1	5	1

* B_1 is Availability, B_2 is Maintenance Requirements, B_3 is Readiness, B_4 is Cost, B_5 is Durability, B_6 is Controllability

When $CR < 0.10$, the consistency of the judgment matrix is considered to be acceptable, otherwise, the judgment matrix should be suitably revised.

After our test, the maximum eigenvalue of the criteria Level matrix is 6.569546, $CR = 0.091862$ and the test is passed.

And the weight vector is:

$$\mathbf{W} = [0.21 \quad 0.03 \quad 0.06 \quad 0.25 \quad 0.11 \quad 0.34]$$

To measure the consistency of the judgment matrix, we calculate the Consistency Index (CI) and the Random Consistency Index (RI). For a judgment matrix B related to factor A , the consistency of each element b_{ij} is evaluated by the consistency ratio CR . The consistency ratio is calculated using the following formula:

$$CR = \frac{\sum_{j=1}^m CI(j)a_j}{\sum_{j=1}^m RI(j)a_j}$$

If $CR < 0.10$, the consistency of the judgment matrix is acceptable.

$$B_1 = \begin{pmatrix} 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 3 & 3 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 1 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 1 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & \frac{1}{3} & \frac{1}{3} & 1 & 7 \\ 3 & 1 & \frac{1}{3} & 5 & 7 \\ 4 & 3 & 1 & 7 & 7 \\ 1 & \frac{1}{5} & \frac{1}{7} & 1 & 3 \\ \frac{1}{7} & \frac{1}{7} & \frac{1}{7} & \frac{1}{3} & 1 \end{pmatrix}, \quad B_3 = \begin{pmatrix} 1 & \frac{1}{2} & 1 & 2 & 5 \\ 2 & 1 & \frac{1}{3} & 3 & 5 \\ 1 & 3 & 1 & 5 & 5 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{5} & 1 & 2 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & 1/2 & 1 \end{pmatrix},$$

$$B_4 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{7} & 2 & 6 \\ 2 & 1 & \frac{1}{5} & 2 & 7 \\ 7 & 5 & 1 & 7 & 9 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{7} & 1 & 8 \\ \frac{1}{6} & \frac{1}{7} & \frac{1}{9} & \frac{1}{8} & 1 \end{pmatrix}, \quad B_5 = \begin{pmatrix} 1 & 2 & 1 & 1 & 6 \\ \frac{1}{2} & 1 & \frac{1}{3} & \frac{1}{3} & 9 \\ 1 & 3 & 1 & 1 & 7 \\ 1 & 3 & 1 & 1 & 5 \\ \frac{1}{6} & \frac{1}{9} & \frac{1}{7} & \frac{1}{5} & 1 \end{pmatrix}, \quad B_6 = \begin{pmatrix} 1 & 2 & 6 & 1 & \frac{1}{2} \\ \frac{1}{2} & 1 & 4 & 2 & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{4} & 1 & 3 & \frac{1}{6} \\ 1 & \frac{1}{2} & \frac{1}{3} & 1 & \frac{1}{4} \\ 2 & 3 & 6 & 4 & 1 \end{pmatrix}.$$

* C_1 is CTD sensors, C_2 is Doppler Velocimeters, C_3 is Multibeam sonar, C_4 is Lifeboats, C_5 is Diving hooks

In conclusion, the Multibeam sonar is the optimal search device show in Table 5.

Table 5: Analysis Results

Scheme	W of c_1	W of c_2	W of c_3	W of c_4	W of c_5
b_1	0.27	0.27	0.27	0.09	0.09
b_2	0.12	0.27	0.49	0.08	0.04
b_3	0.22	0.25	0.39	0.69	0.05
b_4	0.12	0.17	0.59	0.10	0.03
b_5	0.25	0.15	0.29	0.27	0.04
b_6	0.24	0.17	0.09	0.10	0.40
Total sort weight	0.22	0.20	0.30	0.11	0.17

7 Search Path Model

The search area in water is much larger than the single or cluster detection range, so it is necessary to delineate search partitions and plan search paths, which will help improve search efficiency and avoid repeated searches. Target search is a dynamic process that continues to adjust as the search mission progresses, including adjusting the search area, optimizing the search path, detecting underwater terrain and landforms, searching for positioning, and avoiding obstacles and collisions, etc. The search path can be selected based on the search landform characteristics and the shape of the search area, and can be further optimized according to the search task.

7.1 Initial Deployment Point

A key challenge in considering search and rescue operations after a submersible loss is to accurately predict the current position of the submersible. Since a submersible may not remain stationary after a loss, the initial dynamics and the effects of ocean currents may cause the submersible to continue to move. This mobility means that even if the host vessel is able to quickly reach the last known position of the submersible, the submersible may have already drifted to another location due to ocean currents. Considering that the host vessel travels much faster than the speed at which the submersible drifts with the ocean currents, the purpose of this paper is to predict the path of the submersible as it drifts over time in order to determine a more accurate initial deployment point for search and rescue.

7.2 Probability of Occurrence of Submersibles

We use the Monte Carlo method to estimate the probability of a submersible being present at each location due to ocean currents. The Monte Carlo method is a numerical simulation technique based on random sampling and is suitable for solving problems with uncertainty and complexity. Considering that ocean currents can be extremely complex at small scales with many perturbations and disturbances, the Monte Carlo method is able to effectively capture this complexity and variability through the generation of a large number of random samples, thus providing an estimate of the probability distribution of the possible locations of the submersible. For each point in the ocean is sample i , the position is updated for each point in time as:

$$x_{i,t+1} = x_{i,t} + V_{o,t} \cdot \Delta t + C(t, x_{i,t}, y_{i,t}) \cdot \Delta t \quad (44)$$

$$y_{i,t+1} = y_{i,t} + V_{o,t} \cdot \Delta t + C(t, x_{i,t}, y_{i,t}) \cdot \Delta t \quad (45)$$

At time interval Δt , the positions x, y are updated based on the previous positions, V_0 represents the initial velocity, and C is the control variable. At the start of each period, given a pair of initial positions (x_1, y_1, x_2, y_2) , an update function is used to predict the positions at the end of the period. This function can be used to calculate the probability of reaching the final position from the initial positions, denoted as $P(x_1, y_1, x_2, y_2)$:

$$P(x_1, y_1, x_2, y_2)$$

7.3 Search Pattern

7.3.1 Phase 1: Full Search Phase

In the early stages of a submersible crash, it is an intuitive and theoretically effective strategy to use the initial deployment point as the center of the search campaign to maximize the probability of finding the submersible within the first few hours. Therefore, setting a search radius around the initial deployment point and conducting a full search within that radius is an effective initial search strategy.

To effectively set the search radius, the search range can be optimized using an annealing algorithm, a heuristic search method that simulates the process of annealing matter to find an optimal solution. The annealing algorithm helps to progressively refine the search space by allowing to initially jump out of the local optimum and explore a more appropriate search radius, thus increasing the probability of finding the submersible at an early stage.

Through this process, the search radius is gradually optimized until an optimal search range is found. The probability of finding a submersible can be considered as a function of time and search radius, of the form $P(r)$.

7.3.2 Phase 2: Moving Rectangle Search Phase

After completing the initial comprehensive search phase set up based on the annealing algorithm and failing to find the submersible, the paper moves to the second phase. This strategy aims to explore areas where the submersible has drifted further away due to factors such as currents. When confronted with a larger target area, shifting to a rectangular search pattern effectively improves the efficiency of the search coverage, and thus the use of a full search approach is no longer applicable.

7.4 De-weighting

In this study, a rectangular search strategy was used and combined with Monte Carlo algorithm to evaluate the probability of finding a submersible in different search areas to improve the search efficiency. By generating random points within the regions, the Monte Carlo algorithm helps to estimate the probability of finding a submersible in each region. Regions that have been searched and no submersibles were found

had their sample point probability adjusted to zero, while unsearched sample points were dynamically weighted according to the effects of ocean currents to prioritize those regions where submersibles were more likely to be found. This process highlights the dynamic reduction of weights, which is described below:

$$W(i, t) = W_0(i) \exp(-\lambda D(i, t)) \quad (46)$$

7.5 Expanded Matrix Search

We used **expanded matrix search**. The diving search process uses the information of the initial search point and search path of the expanded matrix to minimize repeated visits to the same location in the expanded matrix. It considers how to reduce repeated paths and resource usage to ensure that the search in the expanded matrix becomes the lowest time and resource usage algorithm.

The blue dot in the figure represents the search path, the arrow represents the search direction, and the red star represents the starting point.

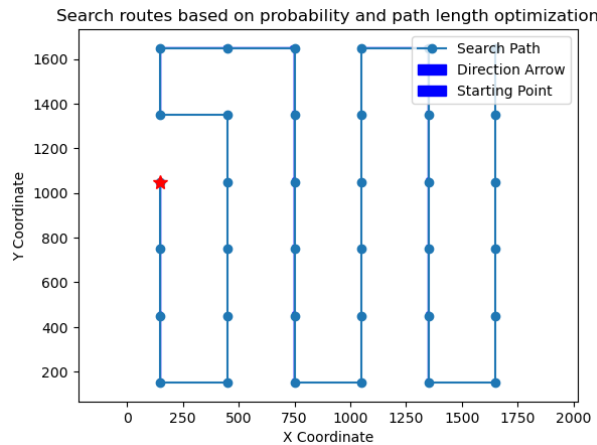


Figure 8: Search Path

8 Generalized Use of The Model

Extending the model to account for other tourist destinations (e.g., the Caribbean), as well as multiple submersibles in the same area, we need to make changes to the way the submersibles operate to adapt to other geographic environments.

8.1 Adaptation to Other Tourist Destinations

Location change model:

$$P(L_{\text{new}} \mid L_{\text{old}}, G) \quad (47)$$

Given a directed graph G , we cannot find a path from any node L_{old} to any other node L_{new} within the graph. That is, for any given starting node in the graph, there is no possible end node that can be reached, which implies that the graph is not strongly connected.

Model adjustments and formula changes:

$$P(R \mid L, E, M, T, S) \quad (48)$$

This formulation is an extension of the original SAR model to include positional information L . At different destinations, changes in geography may affect equipment performance, search patterns, etc., so the model needs to be adjusted accordingly. This may include adaptation of equipment to different sea areas, adjustment of search strategies, etc.

8.2 Multi-submarine Mobile

Position change model (considering velocity and direction):

$$P(L_{\text{new}} \mid L_{\text{old}}, V_i, \Theta_i, G) \quad (49)$$

In a directed graph G with a set of vertices V , if there is a directed edge from any vertex L_{old} to another vertex L_{new} , then the graph G is said to be strongly connected. Strongly connected means that for any pair of vertices in the graph, there is a sequence of directed edges that connects one vertex to the other in the direction of the edges.

Communication cooperation:

$$P(C_{\text{success}} \mid L_i, L_j, P_{\text{comm}}) \quad (50)$$

A directed graph G is said to be strongly connected if, for any vertex pair P_{comm} in G , there is a path from each vertex in the pair to the other. In other words, there is a path in both directions between any two vertices, which makes the entire graph strongly interconnected.

8.3 Overall Assessment

The overall probability of search success is calculated as follows:

$$P(\text{overall}) = \sum_{i=1}^n P(R \mid L_i, E, M, T, S) \cdot P(L_{\text{new}} \mid L_{\text{old}}, V_i, \Theta_i, G) \cdot P(C_{\text{success}} \mid L_i, L_j, P_{\text{comm}}) \quad (51)$$

The directed graph G is strongly connected if for any two vertices in the graph, there is a path from one vertex to the other. The existence of a path from vertex L_{old} to vertex L_{new} can be calculated using the probability of path existence:

Path existence probability (the probability that a path exists from one vertex to another):

$$P(L_{\text{new}} \mid L_{\text{old}}, V_i, \Theta_i, G_d) \quad (52)$$

If the graph does not contain any vertex pairs that can form a path from one to the other, the graph is not strongly connected.

Communication success probability:

$$P(C_{\text{success}} \mid L_i, L_j, P_{\text{comm}}) \quad (53)$$

In a communication model, the communication success probability between any two vertices cannot be zero.

9 Model Evaluation and Further Discussion

9.1 Strengths

Our model has the following strengths:

1. A large amount of literature and knowledge on Olympics has been reviewed and there is a rich background knowledge.
2. The models in this paper were processed and tested before modeling to make the data more real and reliable.
3. The parameters were set very carefully and compared with the actual data many times.
4. We performed sensitivity tests using the data after the model was built.

9.2 Weaknesses and Further Improvements

Our model has the following limitations and associated improvements:

1. If we had more complete data, the analysis of the position of the submersible would be more accurate.
2. The assumption that the trajectory of the submersible can be approximated close to the search route is somewhat idealized. If the trajectory of the submersible is taken into account, more practical models and results can be obtained.
3. Some of the approximations have been applied to modeling other travel locations, which may lead to less than optimal scenarios.
4. AHP has limitations in research problems due to its reliance on subjective factors, inability to rule out inconsistencies, and being too crude for high-precision decision-making problems.

9.3 Sensitivity Test

In section 5.6, 7.5, and 8.3, we performed careful error analysis and sensitivity testing of the model. Through these rigorous assessments, we were able to identify and quantify the level of error present in the model predictions. The results of the tests showed that the errors were very small, indicating that the model was operating with a high degree of accuracy. The sensitivity tests also demonstrated the robustness of the model, confirming that its performance remained consistent across a range of different inputs and conditions. Thus, our model has a high degree of accuracy.

Searching for Submersibles

Our goal is to optimize the deployment and search patterns of submarines, and develop a model that minimizes lost tracking time

Location Prediction Model

In order to develop a more accurate model for predicting the position of a submersible.

Equipment Selection Model

optimize the search efficiency for the submerged wreck while taking cost effectiveness into account.

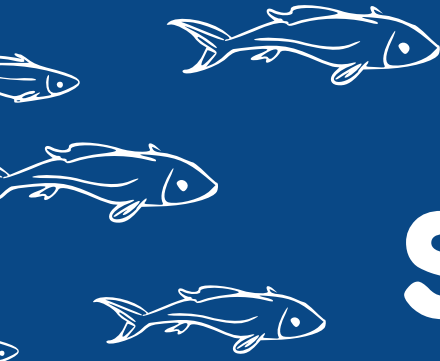
Search Path Model

focused on developing an efficient submersible search procedure

Generalized Model

adapts to localized ocean conditions

It's time to explore deepest parts of the ocean!



SEARCHING SUBMERSIBLES

- Motion model based on the dynamic basis, natural factors and the influence of advanced equipment
 - Search efficiency is optimized by AHP
 - Effectiveness and applicability of the algorithm are effectively demonstrated
- 

OUR MODLES



LOCATION PREDICTION MODEL:

- Uncertainties
- Equipment
- Natural factors
- Mechanical defects
- Error analysis



EQUIPMENT SELECTION MODEL:

- Equipment Selection
- Cost-benefit Analysis



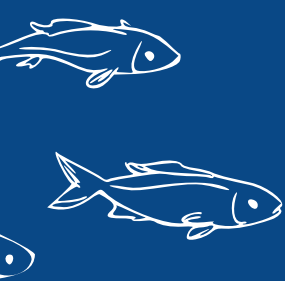
SEARCH PATH MODEL:

- Initial Deployment Point
- Probability of Occurrence of Submersibles .
- Search Pattern
- Phase 1: Full Search Phase
- Phase 2: Moving Rectangle Search
- De-weighting
- Expanded Matrix Search



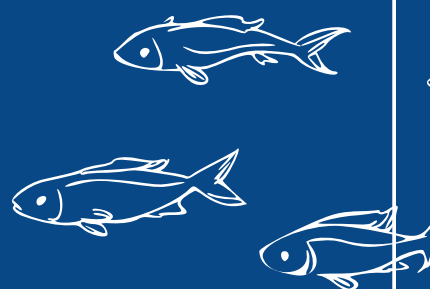
GENERALIZED MODEL:

- Adaptation to Other Tourist Destinations
- Multi-submarine Mobile .
- Overall Assessment



EQUIPMENT

- CTD sensors
- Doppler Velocimeters
- Multibeam sonar
- Lifeboats
- Diving hooks



Multibeam sonar is the optimal search device

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